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Indonesian Transmigration Projects and Living Conditions: Case Study of Built Capital Transformations

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Abstract This study examines a community resettled in Sumatra, Indonesia by the Indonesian Transmigration program, an official government-sponsored resettlement approach of the Suharto era aimed at reducing population pressures in Java and boosting economic development in the outer islands. Since transmigration projects in the 1980s, little research has been conducted on its outcomes. This paper explores the Sitiung Transmigration project, one of the largest transmigration projects in the country, which resettled 3,201 families between 1976-79 from Java to Sitiung in West Sumatra province through the construction of a dam in Wonogiri. People were resettled to land that belonged to indigenous Minangkabau residents. Although the lack of adequate compensation for land takings for the Minangkabau raised the potential conflicts, as has occurred in other transmigration projects, our findings show that the pre-existing Minangkabau community has channeled its resettlement grievances towards the government and the two communities get along well, contributing to successful adaptation. Adaptation was also possible due to the strong social capital of the resettled community, which influenced the physical transformations of the built capital. Findings show that efforts to improve their housing and community labor mobilization were the main factors responsible for improved built capital and living conditions.

Keywords. *Community adaptation, Resettlement, Indonesian Transmigration Projects, Minangkabau, Built capital*

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1 Introduction

In the past 40 years, Southeast Asian countries have suffered from rural-urban migration, accelerated economic growth, and unequal development of urban and rural areas (Terminski, 2015). The increase in density and less access to land and resources increases poverty. In response, some countries have implemented political and administrative strategies to mediate problems caused by densely populated areas and scarcely occupied rural areas. The policy on involuntary resettlement, often by the development induced, has spread successfully across the world and was supported by multilateral lenders such as the World Bank and many corporate finance groups. By considering the environment and other social policies, the lenders aim to manage the project risks and its implementation and monitor the impacts. However, the projects often fail to protect people displaced by the development projects (Price, 2019). The critiques pointed out that material losses can be replaced, but livelihoods and well-being are difficult to ensure (Ariyani et al., 2019), and also working conditions or economic sustainability has improved little (Hay et al., 2019).

The Indonesian transmigration program (ITP), or *Transmigrasi* in Indonesian, is a government-sponsored response to these phenomena. Since its independence from Dutch colonization (1945), the government has been promoting this internal migration and resettlement project based on rural agrarian reform, capital-intensive development, large-scale migration, and intensification of agriculture (Bobsien, 2013; Lucas & Warren, 2013). Transmigration has become a force for rural and regional development, national security, and poverty reduction for landless farmers. It focused on moving low-income and landless populations from dense areas like Java and Bali Island to less populated areas like Papua, Kalimantan, Sulawesi, and Sumatra Island. The ITP resettled almost 2.5 million people between 1979 and 1984 (Benoit & Pain, 1989) and became the world's largest government-sponsored resettlement program. The outer islands of Sumatra, Kalimantan, Sulawesi, and Papua, previously rich in dense forests, resources, and indigenous cultural diversity, suffered severe impacts due to transmigration. This process had enormous environmental and social costs, and as it originally received assistance from multilateral lenders like the World Bank and the Asian Development Bank, it was widely criticized.

Transmigration has meaningfully contributed to national and rural development, helping to extend and diversify agricultural production, and accelerating the development of infrastructure and services. Yet the program has been blamed for mismanagement, with some resettlements abandoned due to their remote location, isolation, or infertile land. Resettlement has also caused deforestation, loss of biodiversity, and environmental degradation of the rainforests and swamp ecosystems, leading to communal conflict over land (Benoit & Pain, 1989). In the face of these developments, many international lenders pulled out of the program by the 1990s, and academic research languished. Some previous studies on government-sponsored resettlement projects in developing countries showed that housing transformation and extensions played an important role in the adaptation process of resettled communities (Carrasco, Ochiai & Okazaki, 2016; O'Brien, 2009; Shirefaw, 1998; Tipple, 1999). However, after many years of successes and failures, many lessons can be learned from the transmigration projects.

Indonesia was a pioneer in internal population relocation. Indonesian transmigration is the world's oldest and largest sponsored internal migration project. With a history spanning more than 80 years, the case of transmigration in Indonesia offers unique insight into the long-term outcomes of resettlement projects, possibilities for the future role of resettlement projects, and possible implications for other countries. Several studies have been published, and the focus of these studies was concentrated on

governmental policy, economic development, or environmental impacts (Arndt, 1983; Gondowarsito, 1990; Hardjono, 1986; Suratman & Guinness, 1977). Unfortunately, most of these were written more than 30 years ago. However, research on the housing and living environment of transmigrants is extremely limited. Pain et al. (1990) is one of the studies that provides analyses of transmigrant dwellers and settlements focusing on physical aspects.

The study site of Sitiung, a mountainous area of Dharmasarya regency, is situated in west Sumatra province, 200 km from Padang city and more than 1,700km from Wonogiri, Central Java. More than 40 years have passed and no study has been made about the outcomes for this largest resettled community. This study fills this gap by investigating the adaptation process and the current situation of Sitiung.

1.1 Objectives

This paper aims to investigate the post-resettlement adaptation of Sitiung resettlers and its impacts on their built capital and living conditions. The study will show how the community's built capital and transformations have improved living conditions. The main objectives of this study are:

1) To understand the current conditions of built capital focused on residential land, basic infrastructure, and housing
2) To understand physical transformations on built capital made by transmigrants to cope with challenges and adapt to the resettlement
3) To analyze the current living conditions of resettlers

To achieve these aims, the following topics were studied: current built capital, current and past living conditions of transmigrants, ownership changes, coping capacity, and adaptation to resettlement.

1.2 Methodology

Built capital and Community Capitals Framework (CCF)

The research applied a Community Capital Framework (CCF) comprising the following Community Capitals: 1) natural; 2) cultural; 3) human; 4) social; 5) financial and 6) built. A capital has been defined as any type of resource or asset capable of producing additional resources. It can be an important indicator of community resilience and adaptation (Fey S., Bregendahl C., & Flora, C., 2006).

This paper will focus on analyzing built capital. Built capital refers to physical assets that support the community, most commonly including sewer and water systems, roads, electronic communications, and housing. It can be understood as the infrastructure facilitating the use of other capitals (Jacobs, 2011). The types of built capital studied were mainly residential plots, basic infrastructure, and housing, which transmigrants originally received from the government as part of their resettlement allotment. Their past and current conditions and transformations performed by transmigrants were the focus of the study. Housing "transformation" can be understood as a phenomenon of spontaneous transformation related to human behavior, defined as "any alteration, addition, extension or modification of a house" (Khan, 2014). The investigation was conducted of the buildings themselves as well as the surroundings, including the residential infrastructure and residential plot.

Physical and measurement survey

Built Capital and physical transformations were surveyed according to Table 1:

Table 1. Built Capital elements

Elements	Items studied	Method of research and analysis
Land	Land use and ownership, redistribution, changes in traditional land tenure and use. Maps, land redistribution, and plan site, land contracts, pictures of old resettlements, and governmental data about land redistribution. Residential plot ownership and changes.	Map making with drawing software, Household (HH) interviews, field surveys, measurement, and local government data.
Infrastructure	access to electricity, water, sewage system, roads and public facilities	HH interview, field survey, measurement, and local government data.
Housing	HH information, housing materials, size, and conditions. Housing drawings and photos.	HH interview, field survey, measurement, photography, drawing.

To understand the adaptation level of the resettlement, living conditions were studied and divided into three main groups: 1) permanent (optimal), 2) semi-permanent, and 3) temporary (minimal). This division includes infrastructure access and housing conditions, showing how housing conditions can reflect the economic situation and adaptation phase of the household. This division was made based on Indonesian standards of construction and building guidelines (Republic of Indonesia, technical guidelines of household development of 2002, building regulation review of 2005, 2016, housing rental regulations 1963).

Sampling

For a deeper understanding of the Transmigration project, key stakeholders involved in the resettlement process were interviewed, including government officials (4), community leaders (3), and transmigrants (102). A literature review concerning post-resettlement outcomes in Indonesia and Development-induced Displacement and Resettlement complemented. The primary data collection was drawn from two surveys (an interview survey and a measurement survey), which were then complemented by in-depth interviews, quantitative data analyses, and field observations, made between August 2015 to December 2017 (see Table 2). Government transmigration documents and quantitative data from government offices in Jakarta and Sumatra were studied. Questionnaire surveys and physical measurements were complemented with field observation.

Table 2. Targeted households and survey contents

Survey	Landholders	HH	Period
Survey 1	69	102	2015.6-2016.8
Attributes: Pre-immigration information, current status of the house, residential plot, and infrastructure access. Housing drawing and map making.			
Survey 2	13	21	2016.8-2017.9
Attributes: Income sources and amounts, residential and cultivation land holdings, family members, housing conditions and living conditions, level of satisfaction (before/after transmigration). Detailed housing transformations, detailed measurement survey with drawings.			

The Household (HH) interviews mainly comprised two distinct groups: transmigrants and spontaneous migrants. Transmigrants are original settlers from 1977. During the first survey in November 2015, the research team measured and collected information on resettlers' property, land, economic status, housing, and access to basic infrastructure (water systems; electricity; roads; telecommunications; fuel, and public facilities). The respondents were selected through intentional sampling (targeting all the houses in one area) and surveyed in person (based on the residential plot and housing characteristics, housing construction materials, and housing maintenance management situation). Information was collected with

household questionnaire surveys (HHQS), interview surveys, field measurements, and quantitative data provided by the local government.

The second survey included semi-structured interviews and measurements in 21 households in 13 housing plots (of which 12 are original transmigrants). In addition to the previous questions, respondents were asked about family structure, land ownership conflicts, lifestyle and living conditions, feelings about new and old living environments, and level of satisfaction. These questions included information about the previous and current conditions (before and after resettlement) for comparison purposes. Data was also collected on population, land use, land distribution, infrastructure, and mapping from official government records. Regarding local Minangkabau people, interviews were conducted interviews with clan representatives (2 out of 3 clans) as well as government officials (Minangkabau), and some non-structured interviews were undertaken with a few residents. The total number of housing and households, landholdings, and housing conditions (size, materials, deteriorations, quality of construction, and structure) were also studied, by using information taken from household questionnaire surveys and housing measurements, drawings, and calculation. This was processed through a drafting software application.

Through this process, land use patterns, infrastructure, and housing transformation were identified and divided into categories for further analysis. Motivations for transformations were also identified from qualitative data from in-depth surveys.

2. Outline of the study site

2.1. Background of Dam-induced Displacement and Sitiung Transmigration Project

Economic growth has led to increased displacement due to infrastructure projects, including dam construction, in Indonesia. Some of the displaced population, either voluntarily or involuntarily, were included in Transmigration-based resettlement schemes, and Sitiung is the largest example of that (Gondowarsito, 1990).

The Sitiung project (1976-1978) was the largest transmigration project carried out by the Indonesian government. Unlike in other transmigration resettlements, in this massive dam-displaced resettlement entire communities were resettled virtually intact (known as *bedol desa* in Indonesia), not usually seen in other Transmigration projects. The construction of the Wonogiri dam (completed in 1981) in central Java affected six sub-districts, moving a total of 10,139 families (41,365 people) between 1976 and 1980. They were forced to relocate to Sitiung 1, Sitiung 2 (both in west Sumatra), and other resettlements in south Sumatra,

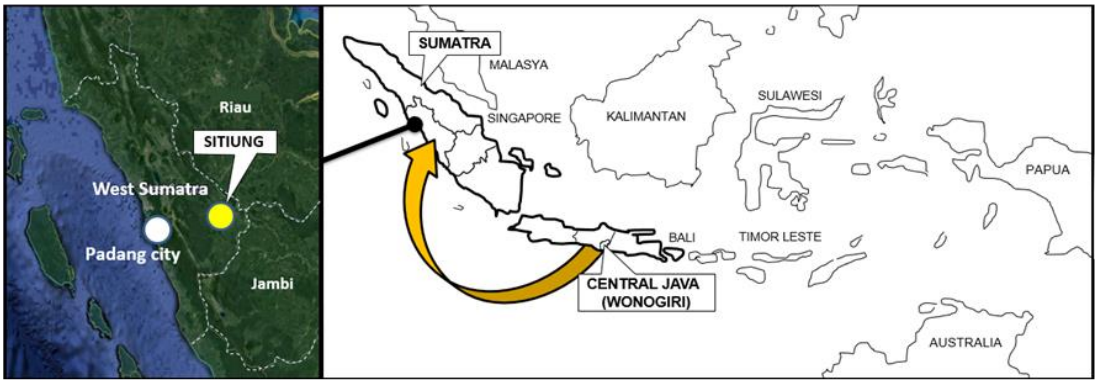


Fig. 1. Location of Wonogiri and Sitiung.

Jambi, and Bengkulu (see Figure 1).

In 1976-77 a total of 2,001 households comprising 8,815 people were relocated from Danuarjo Village, Wonogiri, to Sitiung 1, under the procedures specified by the Indonesian transmigration project guidelines. During 1977-78 4,766 people were relocated to Sitiung 2, comprising a total of 13,581 people. This study focuses mainly on Sitiung 1, since it received the largest number of transmigrants.

Being the first and biggest project of this nature, as well as being sourced in President Suharto's home district of Wonogiri, the Sitiung project received considerable cooperation from government agencies, media attention, and publicity. This experimental attempt to create a new model for resettlement is recognized by some researchers as an exceptional example of multisectoral coordination (Gondowarsito, 1990).

The budget allocated for the project was also higher than in most settlements. In Repelita 1 (Repelita=5 years' national development plan), the estimated cost per household for relocation was Rp350,000, and in 1977 it increased to Rp500,000. In Sitiung, however, the estimated cost was approximately Rp1.3 million or Rp1.8 million including infrastructure (Suratman & Guiness, 1977) .

Intact village resettlement schemes like this one can be advantageous to transmigrants since they come together with friends, family, and neighbors, facilitating the stability of a retained village social structure, greater cohesion, and a smoother adaptation process. However, this can cause problems for integration with indigenous populations and can create tensions with the local culture. Transmigrants were resettled to land that traditionally belonged to indigenous people known as Minangkabau, one of the largest matrilineal ethnic groups in the world. Suratman and Guiness (1977) stressed the fact that transmigrants in Sitiung were being welcomed by local village leaders with respect and mutual cooperation, but with the introduction of 2,000 households to the area, there was the risk that any sense of obligation towards local people could be lost. Although this paper does not focus on the indigenous population, it will present some background information and chronological impacts of resettlement on their culture and land tenure.

2.2. Data collection and surveyed site

The survey site is located in Sitiung 1, block B (see Figure 2). This site was chosen because it was the only

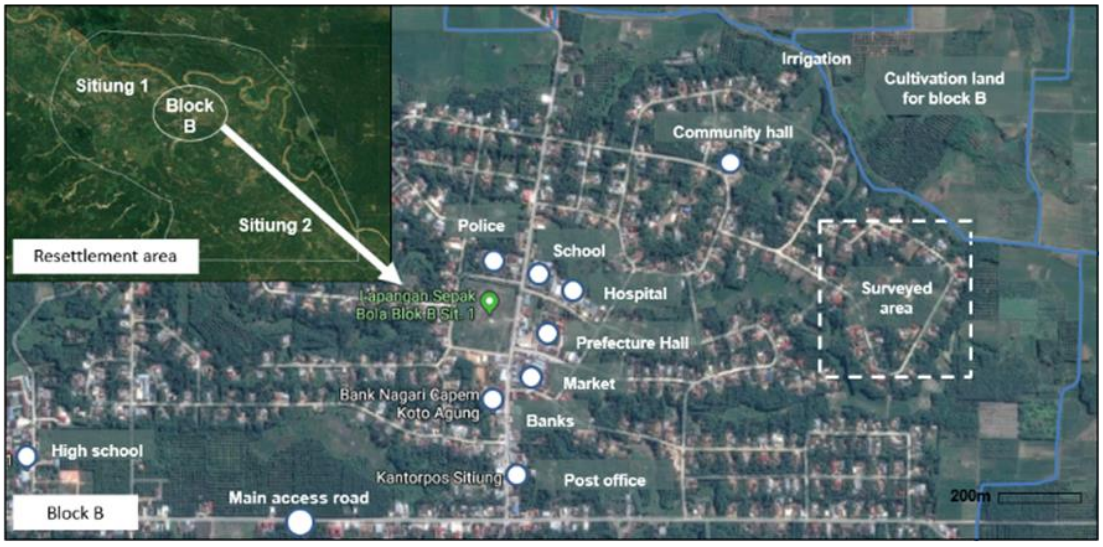


Fig. 2. Resettlement Block B location (*Jorong Koto Agung*) and surroundings. Source: Google Earth, modified by author 2018.

Block of the resettlement that still had the last remaining non-modified original transmigration house and land from the 1977 relocation project. Furthermore, Block B was also the biggest resettled area (565 households), providing a wide sample population. Initial surveys also revealed that this site had households with varied housing and built capital situations. Therefore, this site was selected for the study because it provides an opportunity to compare past and present structures, as well as have a generous and diverse amount of households in comparison to the other blocks.

In total 102 households and 69 landholders were surveyed, of which 59 landholders were original transmigrants from Wonogiri, comprising 10.2% of the 565 original dwellers of block B, and block B comprising 28% of Sitiung 1 original transmigrants. For comparison, intentional sampling included the last remaining non-modified original house of the settlement and the surrounding neighborhood located in RT 3 (neighborhood unit subdivision), as well as the adjacent RT4, 5, and 6 (see Figure 3). The map shows the residential plots where houses are located, and the conditions of internal roads, infrastructure, and common facilities built by the community. The triangle indicates houses that made modifications to accommodate their own business and small shops. House number 2, the last remaining original house from 1977, will be discussed later.

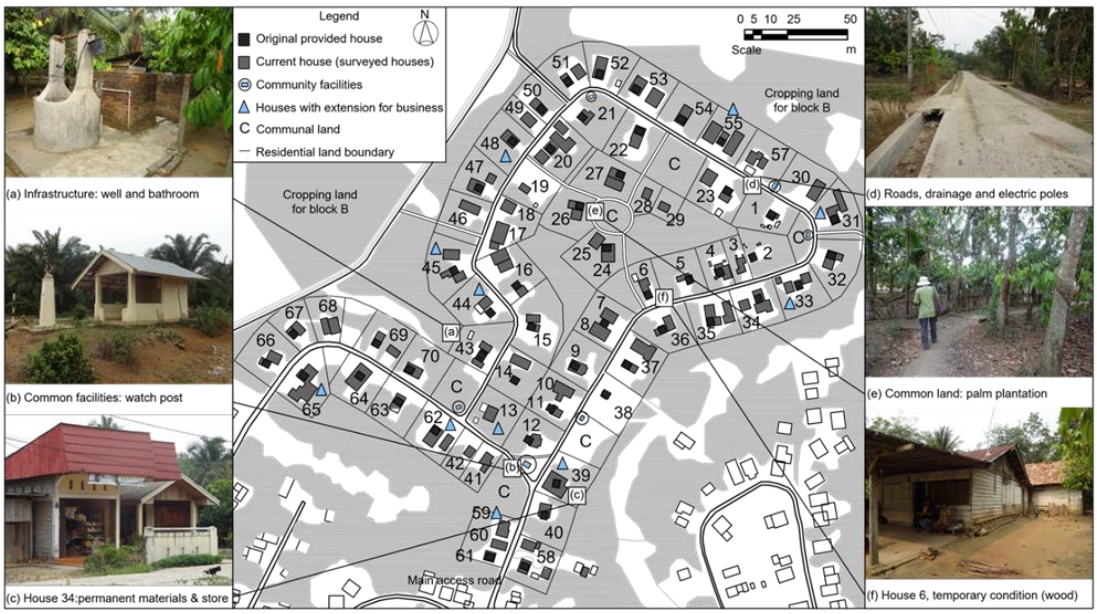


Fig. 3. Block B RT 4,5 and 6: Major infrastructure and current living conditions in the surveyed area.

To ensure the general applicability of the data (blocks A, C, D, and E), we did on-site observation and interviews with leaders and representatives of the areas. By checking the results of the survey against official data made by local governments, it was possible to confirm that all the resettled dwellers were supposed to receive the same house and land and an additional 2 hectares of cultivation land (actual plot sizes varied, as discussed in the next section).

3. Findings

The analytical framework of this paper delves into a comprehensive examination of the factors influencing transmigrat's adaptation, with a focus on Indigenous land and redistribution, changes in

residential plots, changes in infrastructure, changes in housing conditions, the transformation of living conditions, and a summarization of the adaptation process. The analysis also presents an overall dweller's satisfaction with the resettlement process and outcomes.

The study adopts a structured approach, analyzing and presenting these topics from a macro to a micro-scale. For instance, the examination of Indigenous land and redistribution involves a detailed analysis of cropland distribution, incorporating the perspectives of the indigenous dwellers who initially inhabited the area. Changes in residential plots are explored by investigating modifications made by dwellers within their residential boundaries, encompassing actions like division, sale, or acquisition of land as individuals enter or exit the resettlement. Furthermore, the paper assesses infrastructure changes, considering improvements in roads, electricity, access to water, and others. The analysis of changes in housing conditions encompasses factors such as house size, materials used, and overall quality and durability. To facilitate a comparison between households and national standards of living conditions, the infrastructure and housing conditions are categorized based on Indonesian national guidelines, providing insights into the households' living conditions, whether permanent, semi-permanent, or temporary. Finally, the paper offers a nuanced understanding of the overall success of resettlement in enhancing dwellers' living conditions through a comprehensive examination of transformations in living conditions. A graphic timeline is presented, integrating key events influencing adaptation and illustrating relationships between changes in livelihood and built capital, such as land and housing. Qualitative information about dwellers' satisfaction with their current living conditions and the resettlement process is also incorporated, providing a holistic view of the adaptation process.

3.1. Indigenous land and redistribution

The transmigration project in 1977 spurred massive land redistribution at the site by the government. Under President Suharto's New Order transmigration projects scheme, each family joining the program was entitled to receive a 36m² house on a 2,500m² residential plot with two hectares of land for cultivation outside the planned village (Lucas & Warren, 2013; also noted during interviews with households and the local Sitiung government, 2016). Within the specified guidelines regulating transmigration projects, the government was also responsible for land clearing, road construction, irrigation systems for croplands, and later for the construction of institutional buildings such as schools, hospitals, and police stations.

However, in the Sitiung case, the land was originally inhabited by indigenous Minangkabau people. The Minangkabau are one of the world's largest matrilineal societies. In the Minangkabau tradition, the family name, land, and property are passed down from mother to daughter. Prior to the resettlement projects, Sitiung land belonged to 200 households that were divided into three clans: Tigonini, Palian, and Patopan. Traditionally Minangkabau people lived in small villages comprised of 20 to 50 families settling on a roughly 15 km area distributed along the main road. The landscape was comprised of the native jungle, mountains, and cropland used to plant rubber (usually sent to Jambi province, using the river Batanghari to transport it north).

Minangkabau formerly lived in long common houses known as *rumah gadang* (big house), on communal land, known as *Tanah Ulayat*. Indigenous people divided their land under *adat* law, the indigenous land holding law, which was not recognized by the government (Lucas & Warren, 2013). According to *adat*, the clan controlled all local land and resources (rivers, mountains, and forests). This included virgin forests, cultivation land, and housing land.

Minangkabau communities possess unique architecture and cultural value, and some traditional structures have been designated as national historical monuments throughout Sumatra Island. However, many of these structures as well as their culture have been lost, especially in remote locations or places that received transmigrants from different cultural backgrounds. In the surveyed area, Sitiung 1, only two examples of *rumah gadang* remain (see Figure 4).



Fig. 4. Examples of remaining traditional Minangkabau houses.

Property relationships of the Minangkabau and the Javanese are very different. In *adat* indigenous law, according to Benda-Beckmann F/K, property relationships are a result of social and political status, where the dominant property categories are defined by relation to the property holders. Being a matrilineal society, Minangkabau land was owned communally and was cultivated together by local residents where everyone could cultivate and depend on the land for sustenance (Benda-Beckmann F. & Benda-Beckman K., 2006; Beanda-Beckmann, 2013).

By contrast, the Transmigration land distribution is based on the patriarchal Javanese culture, where land division resembles cluster patterns where the land is delimited by well-defined boundaries of ownership. These boundaries are set according to property limits for each nuclear family and extended family, resembling the patriarchal model used by the centralized Javanese government for land redistribution.

These two features of the Sitiung project resulted in friction and led to the government appropriation of communal land. In addition, the allotment of two hectares of cropland promised by the government to transmigrants was not fulfilled. The Minangkabau was supposed to receive the same amount of cropland, according to government promises made during negotiations before the project started (personal communication from local community leaders and local government).

Several Minangkabau reported that promises were not fulfilled and they didn't receive any land. Traditionally, local people used to plant rubber as their main source of income, exporting the rubber to other provinces. In exchange for their land, they reported that the government paid them paltry compensation for rubber plantations (30 cents per tree) which was insufficient to cover their economic losses.

Figure 5 below shows how traditional land ownership was drastically reshaped by the resettlement project. Today there is no more communal traditional land found in Sitiung. The image on the left shows the location of the original Minangkabau settlements and land distribution (in black); it was made according to information from the head of the clans during an interview. On the right is the map of land distribution of the transmigration project according to the government. This map was drawn based on a hand-drawn map from Sitiung district, the last remaining copy of land division plans. The original map was last updated in 2006 by the government, and by that time there was still land that was not certified or legalized. According to the planned map, until 2006, the amount of land designated as belonging to original

indigenous inhabitants was only 42 ha (1.8% of cultivation land), while Javanese transmigrants held the majority of the land with 2299 hectares (98.2% of cultivation land).

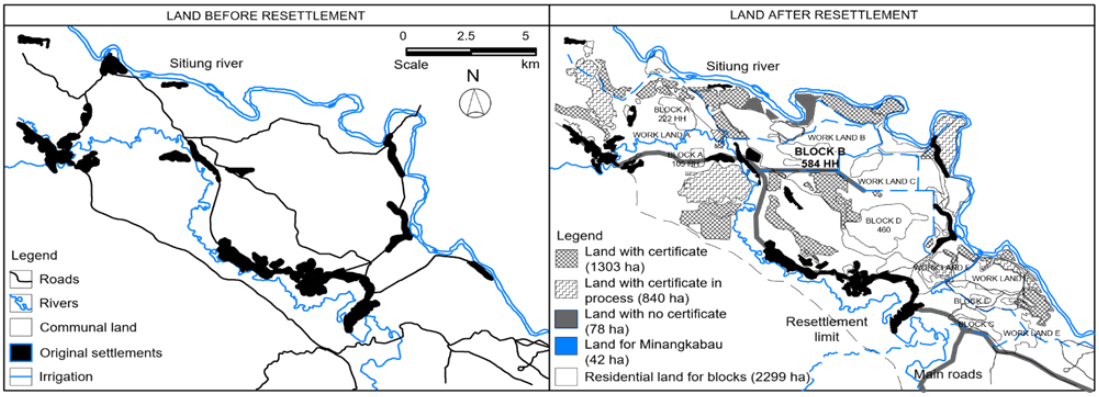


Fig. 5. Map of land distribution of resettled area. In black is land belonging to Minangkabau people. The map on the right show government plans for land redistribution, last updated in 2006 by the local government.

It is not surprising that this land redistribution caused conflict. During the interviews, transmigrant households often said there was no conflict with locals, but Minangkabau representatives emphasized that there had been many conflicts over land ownership, which were often addressed to the government directly, and not to the transmigrants.

The clan representatives said the majority of the indigenous inhabitants did not hold any cultivation land, and that they themselves only had residential plots measuring about 2,500m². Leaders said they were under great pressure from the government, which claims governance rights to all land in Indonesia. The basic agrarian law in Indonesia states that: "the land, water, and atmosphere, including natural resources within them, are controlled by the state at its highest level, as the organizational authority of all people" (BAL/UUPA 5/1960, N2.1), putting the central government above indigenous land ownership. Since Indonesian law has rarely recognized indigenous *adat* law, indigenous land lacking formal tenure is vulnerable to seizure by the state under eminent domain without proper compensation.

Because of these issues with land redistribution, HH survey data shows that transmigrant landholdings in Sitiung average 1.25 ha per household, while they should have received 2 ha. Land redistribution is still in process, although residents do not expect to receive any more land, since most of them received some kind of compensation earlier (in most cases they received a cow in exchange for the remaining land that had been promised to them). As a result of the transmigration way of land distribution and usage, the indigenous communal way of living, including land usage, as well as the collective housing *rumah gadang* does not exist anymore.

It is natural to expect conflict when resettling transmigrants to indigenous land, especially in the case of inequitable land takings and poor compensation, as occurred in Sitiung. Nevertheless, interviewees stressed the fact that there was little conflict between indigenous people and transmigrants. In fact, Minangkabau greatly facilitated the adaptation process of the newcomers by helping them with moving, and by providing jobs and food assistance during their difficult first years. A Minangkabau leader explained that “[the transmigrants] were in a difficult situation, losing their land and moving to an unknown place, so we wanted to help.” Their cooperation was seen by many transmigrants as essential to sustain the resettlement and contribute to the community’s successful adaptation.

3.2. Changes in residential plots

In addition to the cropping land, transmigrants received residential land of 50x50m, as shown in Fig.2. Sitiung suffered from serious floods, especially during the rainy season, and dwellers said that every time a big rain came, the residential land and house would get flooded. Therefore, the first modification dwellers performed was to mitigate the flooding problem, such as elevating and paving the floor and building a drainage system.

Not all the transmigrants could adapt. Household interview data revealed that during the first 20 years, approximately 10% of the transmigrants gave up on the resettlement and returned to Java or migrated elsewhere, selling their land (see Table 3). This is a relatively low percentage, but some transmigrants noted that since their original communities in Wonogiri had been inundated they had nowhere to which they could



Fig. 6. Types of residential plot division, n=102.

return. Despite land issues and chronic flooding, dwellers adapted and the resettlement achieved increased stability and well-being. Later, spontaneous migrants started to settle in Sitiung in search of low-priced land and the opportunity for a better life.

Table 3. Ownership of lots, division/buying and selling land

Type of residential plot	Quantity (%)
0.25 ha. original resettled land (no change)	21 (30.5%)
0.25 ha. original land with new owner (buy/sell)	4 (5.8%)
0.25 ha. original land with several HH (same family)	23 (33.3%)
Partial land (<0.25 ha.) (buy/sell)	19 (27.5%)
HH living in common land (not official)	2 (2.9%)
Total (original and spontaneous migrants)	69 (100%)

Today, many dwellers are spontaneous migrants (representing 23% of the total population), and residential plot transformations to accommodate them represent 17.5% of total modifications of residential plots, including dividing land to sell, renting land and houses, or opening new land (see figure 6).

In summary, of the total 59 original transmigrant landholders, six owners (10%) left the site and sold their land (four sold their entire landholdings, two sold part of their land first, and the rest after some years), 1 owner passed away and 52 original inhabitants still remain (with 10% of those selling part of their land).

A majority (48%) of spontaneous migrants come from Wonogiri. During interviews, they explained that they heard from transmigrant friends and neighbors who had resettled in 1977 that " in Sitiung, there's more opportunity and land, and life is better than in Wonogiri." This was an important motivation for them to move.

Access to land was a major pull factor for spontaneous migrants. Many said they were searching for a better life; for available and cheap land to buy and on which to build their houses. Some came from other places in Java, while some Minangkabau people from outside of Sitiung were included in this demographic.

The majority of land holdings belong to original transmigrants and their children, with an average household residential plot of 1,495m² (0.1495 hectares). Since many of the transmigrants divided their land to sell to spontaneous migrants, they hold on average 1,000m² less than the original allotment, while spontaneous migrants hold an average of 1,093 m² (see figure 7).

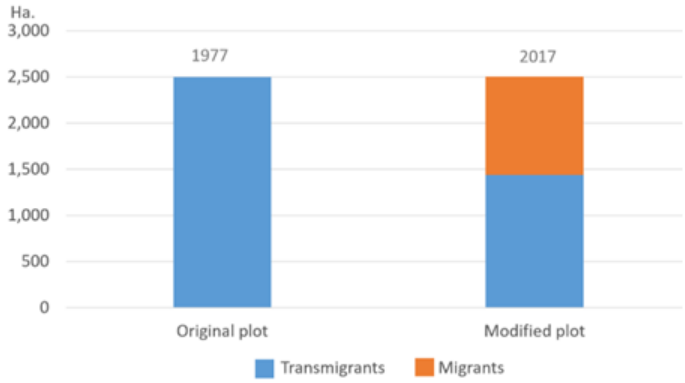


Fig. 7. Residential plot holdings, n=102.

Interviews revealed that motivations to sell the land were various, but mainly the residents needed money to make improvements on their own houses, or to address critical economic needs during the first years. Today, residential landholdings of residents can vary from 500m² to 2,500m².

Selling part of their land provided residents with some extra income, but it wasn't the main activity that contributed to allowing residents to improve their built capital and living conditions. After the first five years of hardships, the increased income due to the diversification of crops and growing tree crops like oil palm and rubber, together with community building practices, allowed for reconstruction, which will be explained in the following sections.

3.3. Changes in infrastructure

For the new settlement, the government provided basic infrastructure: oil lamps for lighting and one well for four families. Of all the problems faced during the first years of adaptation, the most difficult, as described by respondents, can be attributed to the lack of infrastructure (85%). There were no roads, access to water, or electricity, and it was difficult to procure oil for lamps and other purposes. After some years, the government provided electricity and institutional infrastructure such as schools, governmental buildings, and health care centers, but the rest of the infrastructure related to living conditions was built by the transmigrants themselves.

As a cohesive community with strong existing social networks and traditions of community labor exchange (known in Indonesian as *gotong royong*), the residents started to develop improved infrastructure by themselves instead of waiting for the government or external assistance. One of the first modifications they made, in 1979, was improving rainwater collection and canalization along the roads while upgrading road conditions. This was an emergency measure since, according to several residents, they suffered from damaged homes from chronic flooding each year during the rainy season

The community also built common facilities such as village huts (for socializing and security patrols), and storage facilities to store common materials for ceremonies like weddings or festivals. Access to water was also very scarce, as there was only one well for every four families; therefore, making wells for themselves and their neighbors was a priority during the first years.

According to previous research, the irrigation system in Sitiung was built by the farmers themselves (Hoffmann, I., 1991). When transmigrants arrived in 1977, the irrigation system was not yet completed, so many residents began building infrastructure by themselves. According to survey respondents, some of these projects were *gotong royong*, but some residents were paid low wages by the government to build irrigation. International agencies also contributed: One of the early irrigation projects in Sititung was funded by the Japanese government (World Bank, 1979).

According to the Republic of Indonesia Technical Guidelines for Household Development 2002 and the Republic of Indonesia Construction Guidelines of 1963, infrastructure can be categorized as follows, with the percentage of housing in each category shown in Figure 8.

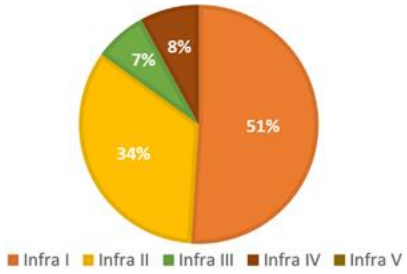


Fig. 8. Infrastructure categories, n=102.

- Category I (maximum): piped water, gas, septic tank, paved street, telecommunications, electricity, W.C., kitchen and bathroom.
- Category II (good): piped well, septic tank, paved street, electricity, W.C., kitchen and bathroom.
- Category III (medium): gas, paved street, electricity, W.C., and kitchen
- Category IV (poor-minimum): oil fuel, electricity, and kitchen
- Category V (none): no presence of the above-mentioned infrastructure elements

During the first years, the community was classified as category V, without electricity but using oil lamps and lacking roads or sewage systems. After transformations, today, 100% of households have access to electricity, 84% have access to their own piped water well, 92% have access to paved roads, 93% have a septic tank, 78% have access to telecommunications and 72% have bottled gas. HH in categories I and II represent 85% of the total, with access to good quality infrastructure comprising almost all the elements presented above, and there is no household in the poor infrastructure category. In addition, chronic flooding problems have been mitigated.

3.4. Changes in housing conditions

The house provided by the government was a wooden house with a roof made of thin corrugated iron, measuring 36m², containing a living room, two bedrooms, and a kitchen. Some of the houses were provided with a well and improvised wooden toilet (see Figure 9). Respondents described the main sources of dissatisfaction with the original house as: high temperatures inside and poor ventilation; primitive earth floor that became easily inundated during heavy rains since the house was not elevated above ground level; and small in size so unsuitable for families. The transmigrants improved and extended their houses over time to match their needs and improve their living conditions.

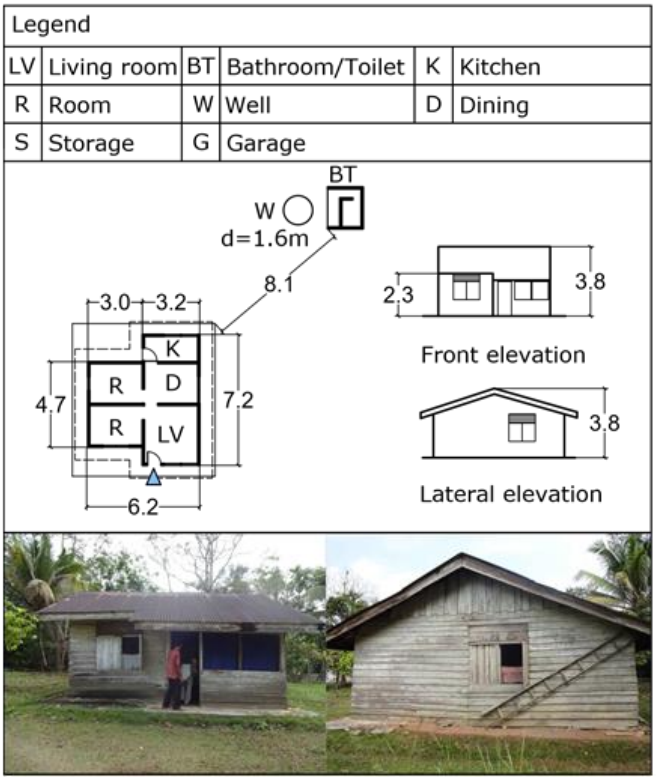


Fig. 9. Original house from 1977. Floor plan, section and elevations.

The first modifications were to elevate the floor and build drainage to cope with floods. Residents also extended the back part of their house to have a bigger kitchen. Since their only source of cooking fuel was firewood, the residents built an exterior kitchen in the back of the house, which is still used nowadays as a supplementary kitchen. The first extensions were made using wood, but later on, they added more rooms or built a completely new house if they had suitable financial resources and could buy better materials. Modifications made during the first two decades were largely performed by the transmigrants themselves with help from the community, while in the past two decades, residents have tended to pay construction workers to build their houses and improve the quality of the house and structure.

Key findings about housing transformation showed that house size increased from an average of 36m² to 130m² to accommodate family needs. In fact, 82% of the houses were completely reconstructed, and only one original house remains intact.

To understand the actual conditions of the house and determine house durability, data collection focused on measuring structure, partitions, and roof materials. Most houses now are made of confined structure materials (64%), often confined concrete blocks, or a mix of blocks, brick, and wood (19%), and

only 18% of the houses still contain elements of the original wooden structures. Inside the houses, 41% of the partitions are made using concrete blocks, 19% with wood, 2% with brick, and the remaining 58% by a mixture of these materials. Corrugated iron represents 59% of the materials used for the roof and tile 10%, while 31% is a mix of these two materials.

Housing durability was classified into three (according to Republic of Indonesia technical guidelines for household development 2002 and Indonesian construction guidelines 2005 and 2016). Durability depends on materials, life expectancy, and type of structure, and it is divided into the main groups:

- Durability I: life expectancy of 20 years, confined structure, materials of brick or concrete block, roofs made with tiles or corrugated iron, floor made with cement, and plastered walls.
- Durability II: life expectancy of 5-10 years, non-confined structure, mixed materials (half brick or block, half wood), roofs made with tiles, and floors made with cement.
- Durability 3: life expectancy of 5 years, wood or bamboo structure, material of wood or bamboo, roofs made with tiles or thatch, and floor made with cement or earth.

It was found that in the case of Sitiung, the house's durability conforms to the structure type. Houses in the durability I category represent 64%, durability II 19%, and durability III houses represent 18%. With the housing durability and infrastructure category, it was possible to calculate the living conditions (permanent, temporary, or semi-permanent) that will be explained in the next section.

Figure 10 shows some examples of detailed measurements of modifications performed by

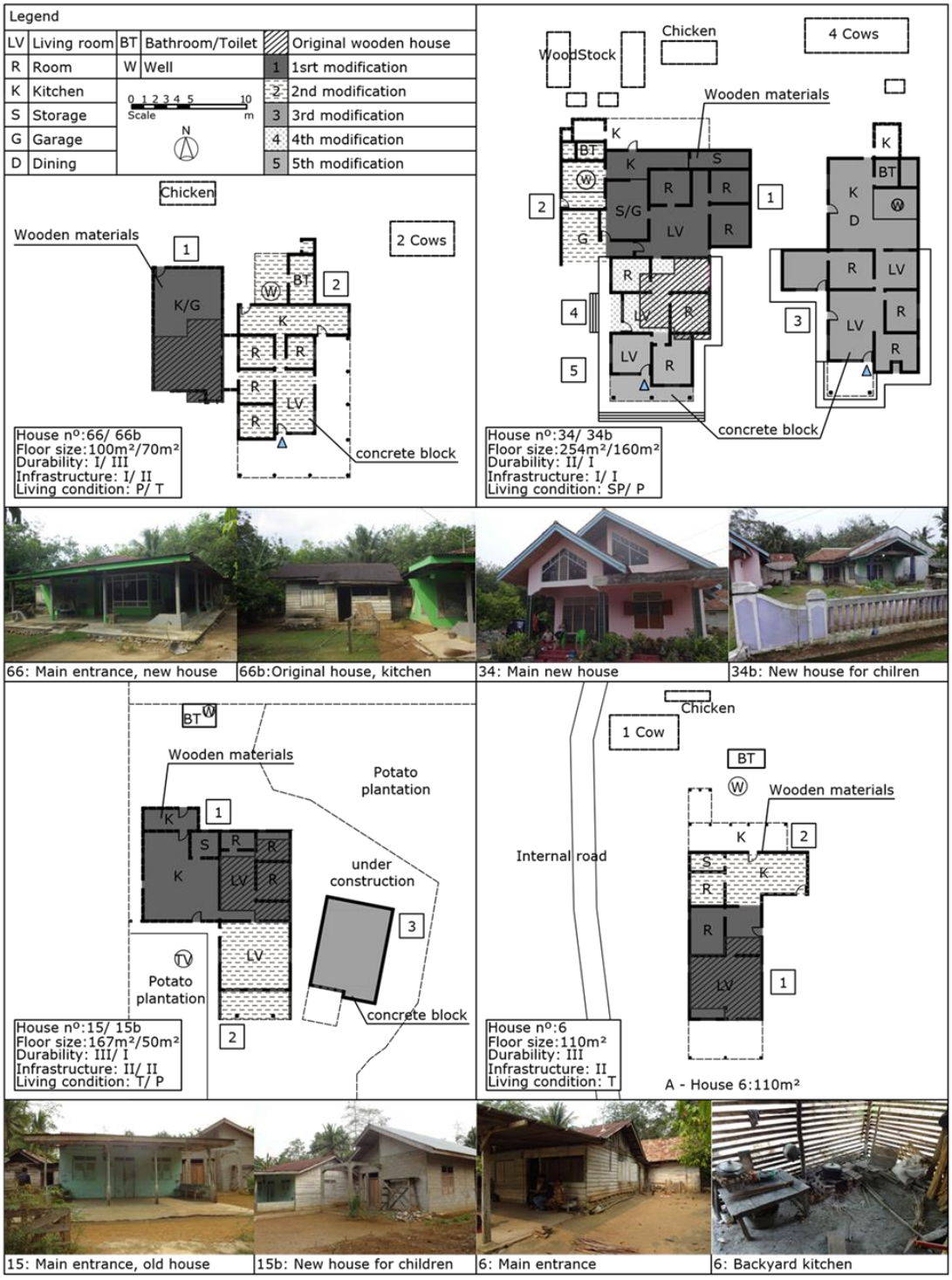


Fig. 10. Examples of housing conditions and transformations.

different families. These four examples, out of 102 houses measured, represent different conditions, different durability, and different types of materials and extensions. By comparing the location and size of the original house with the actual floor plan, we can see how they changed their houses to fit family needs. Few houses kept the structure or facade of the original house, using it as storage, garage, or secondary kitchen. While most of the families constructed completely new concrete block houses, few families still live in wooden houses similar to the original ones.

House 66 and 66b exemplify one example of how the transmigrants built a separate new structure for their new house and used the originally provided house as storage or a secondary kitchen. In this example, the original house was extended to fit family needs during the first years of resettlement (before the 1990s), but it was no longer used for housing as soon as the family had the financial wherewithal to build a new house made of concrete block.

In the case of houses 34 and 34b, the original house was extended in 1992 to meet the needs of a growing family, and the residents continued making modifications, building a new house for their children in 2000 until no trace of the original house remains. This case shows how housing that is built completely new (34b) usually uses one type of material (in this case concrete block), improving the durability of the house and living conditions. Meanwhile, houses that experienced several steps in transformation (34) used mixed materials and structure types, therefore the durability category and living conditions differ for each example. This type of house using mixed materials was very common in Sitiung and represents 19% of the houses that were identified as durability II. This family is an example of a successful adaptation, as improved access to financing enabled them to improve construction with better materials.

Houses 15 and 15b are examples of modifications using wooden materials followed by the construction of a new concrete block house. This is an example of a family that is still adapting from temporary living conditions to more permanent living conditions, with the new house still under construction. Other examples like this can be found in the surveyed area.

Lastly, house number 6 is an example of a family that still lives in very similar conditions to those that first prevailed. Although the house was extended, the materials are fragile and old, and the structure is weak and poorly engineered. This exemplifies a house in category III, which represents 18% of the total. Lack of economic resources is the main reason for families not being able to improve their houses. This family in particular noted during their survey that the crop land they received for cultivation had a flooding problem due to its proximity to the river. Due to chronic flooding and a lack of knowledge of how to improve their crops, they suffer from a low income and have been unable to improve their economic conditions since resettlement. They are frequently helped by neighbors who provide food supplies. This shows how original cropland quality and distribution can directly affect residents' continuing living conditions and housing quality.

Although the outcomes of the transformations performed by dwellers differ a great deal, varying from small houses with one unit to large houses with three or sometimes four houses inside one plot, in the examples shown here one can find some notable similarities in the transformation process. When observing the housing plans, it is evident that for the initial modifications made by residents during the first 10 years, they concentrated on renovating and extending the original house. These transformations were made, first, to cope with the flooding problem; therefore, the first modifications tended to involve elevating and cementing the floor to protect against floods. In the next decade, most of the transformations were to construct new houses or to sell land, especially when spontaneous migrants began to enter the community in larger numbers. From 1997-2007, residents focused on demolishing the original house and building completely new ones, and building additional new houses. This may have been thanks to the extra income generated by selling land and better conditions of adaptation.

Finally, the focus of the last decade was mainly on continuing the demolition of the old houses to build new ones. Due to this, few traces of the original houses can be found in Sitiung today.

After the first phase which was common to all residents, modifications varied: continuing expanding the original house to add a bigger living room and extra rooms but still using wood, building a completely new and separate house with better materials, extending the original house with better materials

and using the original wooden portion for supplementary kitchen and storage, or completely demolishing the original house and building new structures instead. It is possible to find parts of the original house after transformations in only 17% of the houses.

The first modifications were built by residents themselves, with the help of neighbors (as part of *gotong royong*) and sometimes by paid local builders/carpenters. Residents and neighbors continued playing important roles in the modifications, but the last modifications were mainly made by paid construction workers. This helped to improve the house's quality and structure.

Income and expenditures in transformations were roughly measured since many dwellers could not remember how much they spent or were unable to accurately estimate income, which fluctuated by season and year. With data from 21 households taken from survey 2, it was found that they spent an average of Rp66,555,555 (around USD5,000) for the transformations, ranging from small extensions to building new houses (ranging from Rp8,000,000 to Rp125,000,000). The average monthly income of residents is Rp1,772,220 (ranging from Rp400,000 to Rp7,000,000). This shows that households made transformations regardless of their income, although income could influence the size, material, and quality of transformations.

3.5. Living conditions

Living conditions can be analyzed in many ways, usually taking into consideration housing conditions, size, sanitation, ventilation, indoor climate, safety, access to lifelines, and others (Republic of Indonesia 2002, 2005,2016.) Housing conditions and safety were studied according to materials and durability (already explained before), with the premise that the durability of a house correlates with safety and quality. When it comes to size, it was observed that some houses are quite large, with a total size of more than 100m², but may still be constructed of non-durable wooden materials (as in the case of house 6 in figure 10). Therefore, size was discarded and the other components were taken into consideration for the calculation of the living conditions of residents. To calculate living conditions for each house, the infrastructure and house durability categories calculated before were combined. For easier understanding, living conditions were categorized according to the guidelines of Indonesian standards explained in Section 4. It considers housing materials (floor, walls, and roof, durability of the house, and access to infrastructure and its components) and identifies three main categories of living conditions:

- Permanent: house with durability I and infrastructure category I or II
- Semi-permanent: house with durability II and infrastructure category II or III
- Temporary: house with durability III and infrastructure category V or IV

After all the transformations, dwellers improved their living conditions, and now 59% of residents were found to conform to permanent living conditions (with good access to infrastructure and high durability of the houses), 16% to semi-permanent conditions (with good or medium infrastructure and medium durability of the houses) and 25% to temporary conditions (mainly due to houses still built with wood and a few residents with poor infrastructure). However, all residents showed improvements to some extent, and there are no residents living with the same conditions as in the initial phase of resettlement (see Figure 11).

Detailed interviews with residents revealed that transformations are related to their adaptation process and their ability to cope with challenges and adapt. Those who weathered the first harsh years of resettlement and improved their living conditions made great modifications in their house and surroundings.

These modifications helped them to cope with the challenges of resettlement physically and emotionally, expressed by housing extensions, improved materials, improved aesthetics and comfort, small businesses, and others. These achievements in transforming their surroundings represented stability for their present and hope for their future.

Economic factors were important to support these modifications. Many residents improved their economic status by shifting crop production from rice to rubber and oil palm plantations. Several could also increase their production by buying land from those who left the site. In general, economic conditions became better than in Wonogiri by the 1990s, since many could gain more agricultural income in Sitiung. In Wonogiri, farmers lost their cropland and haven't been able to acquire additional land that became more expensive. In addition, since they tend to grow rice and other staples they haven't benefitted from the rising commodity prices of rubber or palm oil.

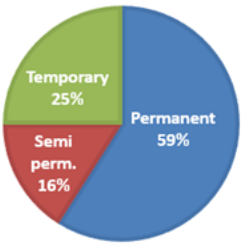


Fig. 11. Living conditions in Sitiung, n=102.

3.6. Adaptation process and dweller's satisfaction

According to the findings, a timeline was drawn for a better understanding of the adaptation process in Sitiung (see Figure 12). Despite the hardships, many spontaneous transmigrants moved to Sitiung (especially from Wonogiri) within the first 10 years of resettlement (1977-1987) searching for better living conditions.

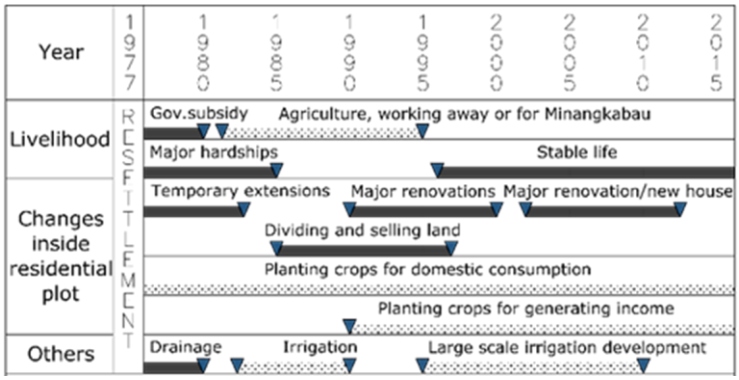


Fig. 12. Overall key events of adaptation process of resettlement from 1977 to 2015, concerning quantifiable and measurable elements that influenced the built capital. Qualitative aspects of the adaptation process such as *gotong royong* are discussed in the text.

Plot land transformations were important to accommodate those migrants and also to help transmigrants accumulate capital and be able to improve their living conditions. Those modifications were made mainly during 1987-1997, showing that resettlement took at least 10 years to achieve a certain level

of stability and be attractive enough to draw spontaneous migrants.

The extra income generated by selling residential plots to spontaneous migrants helped to generate income that was used by transmigrants to build newer and better houses and renovate their old houses. Ownership of cropland was also a factor that changed the economic status of residents, allowing them to improve their living conditions. By planting fruit trees or other crops on their residential plots dwellers were also able to maximize their income, especially with oil palm trees. After the 1990s the irrigation system was in place, so crop production became more stable, and mitigation of floods allowed for better housing and agricultural conditions. After 2000, the price of palm oil went up and they could enjoy higher levels of income.

Not only access to land but also land quality influenced eventual outcomes. Most of the families that are now living in temporary conditions link their current status with the fact that the land they received had poor fertility so they couldn't grow crops successfully. Therefore, equitable distribution and land quality were shown to be an important element in the adaptation process.

Regarding general life satisfaction, it was surprising that despite all the initial hardships, 19 out of 21 dwellers interviewed during the semi-structured survey said they were satisfied with the resettlement project and with their lives. The first reason, many said, is because they received land. The transmigrants resettled from Wonogiri to Sitiung were those who originally did not possess much land or property. By receiving land, they were able to cultivate the land and with their efforts to gain more income and have a better house and hope for a better future.

The second reason for satisfaction was that they were able to come together as a community. One respondent said, "Everyone was my neighbor before and they still are here. I know everyone, we went through hard times but we were together". This shows how the sense of community is very strong and plays an important role in the adaptation of these transmigrants.

4. Conclusion

The Sitiung Transmigration resettlement was a complex and multi-faceted project, and many factors played important roles in the adaptation process. During the first 20 years, 10% of the transmigrants gave up on the resettlement and went back to Java or tried a better life somewhere else, selling their land to spontaneous migrants, causing changes in the residential population. Access to land was the main pull factor for new migrants, and today, they represent 23% of the total population of the surveyed area, and 48% of them come from Wonogiri, the original district of the transmigrants.

The original resettlement housing was extended and now the average house measures 130 m². Modifications were made to accommodate family needs, improve living conditions, and survive floods. Of the total original housing stock, 82% of the houses were completely reconstructed, and only one original house remains. As for building materials, 64% of the houses were built as confined structures using brick or concrete blocks, and 18% are still made of wood. Regarding infrastructure, 100% of the households have access to electricity, 84% have access to their own piped water well, 92% have access to paved roads, 93% have a septic tank, 78% have access to telecommunications and 72% have natural gas.

The physical transformations helped to improve resettlement living conditions. From initial temporary conditions of resettlement, today 59% of the residents are found in permanent living conditions (with good access to infrastructure and high durability of the houses), 16% are in semi-permanent conditions (with good or medium infrastructure and medium durability of the houses) and 25% in temporary conditions (mainly due to houses still built with wood and a few residents with poor infrastructure). Most

of the transformations were performed by the residents themselves and the majority by the community, while government assistance played a minimal role in this process.

Despite many initial hardships, dwellers said they were satisfied with the resettlement project and with their lives. The main reasons were that they received land and also came together as an entire community, thus easing adaptation and allowing them to overcome hardships. Successful adaptation also was possible due to the positive mutual relations and support received from the indigenous local Minangkabau residents, who facilitated adaptation for the new residents by supplying food and labor opportunities. Strong social capital and improvements in their economic conditions after the plantation shift in the 1990s helped residents improve their built capital and living conditions. The adaptation of transmigrants in Sitiung was a complex combination of events and adaptation processes. This study focused on the physical aspects of the built capital and some perspectives from transmigrants about their initial hardships and living conditions today. Future studies should also survey indigenous residents, to study their lived experiences and adaptation process as well.

5. Further discussion and research topics

Although the resettlement took place in an area already settled by indigenous residents, the government approach did not take into consideration local traditional culture and communal land practices, threatening traditional land holdings and structures, which were nearly eliminated after resettlement. Much of the traditional culture was affected and this is a problem for the national and local heritage. Traditional communal land practices no longer exist, and the relationship with the land and Minangkabau lifestyle completely changed to adapt to the ITP resettlement style. Traditional people lost their productive land, cultural landscape, and much more. Conflicts between indigenous people and the transmigration project emerged mainly due to the lack of infrastructure provided by the government during resettlement, inequitable compensation or land distribution, lack of transparency, and unfair access to resettlement benefits. In the case of Sitiung, indigenous people directed their land conflicts towards the government, and a good relationship was maintained with transmigrants. Serious attention must be paid to the development of infrastructure and physical structures of resettlement, as well as applying principles of economic and social equity that benefit the indigenous people as much as the transmigrants.

One of the main lessons of this study is that although housing and land were provided by the government, the improvements in the resettlement were actually the outcome of the long-term activities and transformations performed by the residents and communities themselves. Importantly, physical transformations performed by residents became a key factor for positive adaptation and improvement of living conditions. Also, community-based labor mobilization activities played an important role in many of the adaptation processes, including infrastructure, housing, and coping with flooding. Having pre-existing social networks and relations from before settlement, and being settled intact, were characteristics of the resettlement that contributed to strengthening community bonds and facilitating community mobilization.

More attention should be paid to the quality of the built capital. By learning about their transformations and housing conditions, we can understand more about the adaptation process of resettlement. A study of how residents and communities working together made built capital transformations is a productive methodology for understanding the process of resettlement. Consideration of local conditions of resettlement should be taken more seriously. Identifying patterns in transmigrants' physical needs to sustain their daily activities, family structures, and livelihoods, under conditions of the local environment, is critical for the provision of suitable resettlement and provision of locally sensitive solutions.

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